

ANNABA, Algeria

WMO: 603600

Lat: 36.8220N

Lon: 7.8036E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	39.3	41.2	33.6	28.3	49.5	35.9	30.9	49.1	30.0	54.7	25.6	53.4	7.6	220	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.4	95.1	73.0	91.2	73.3	87.9	73.8	79.1	85.6	77.7	84.6	76.4	83.6	11.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	141.6	83.1	75.5	133.5	81.9	74.1	127.5	81.1	42.6	86.2	41.0	84.7	39.8	83.6	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.0	20.5	18.4	DB	35.0	106.8	2.3	3.5	33.3	109.4	32.0	111.4	30.7	113.4	29.1	115.9
			WB	34.0	81.3	2.1	1.9	32.5	82.7	31.2	83.8	30.0	84.8	28.5	86.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	53.2	53.0	56.4	60.1	65.7	72.3	77.2	78.7	74.5	69.3	61.0	55.3	(6)
	DBStd	10.09	4.00	4.47	4.55	4.34	4.54	4.61	3.84	4.12	4.48	4.83	4.89	4.25	(7)
	HDD50	44	14	18	4	0	0	0	0	0	0	0	1	7	(8)
	HDD65	1628	365	337	271	158	43	2	0	0	0	13	137	302	(9)
	CDD50	5445	114	102	201	303	487	668	844	889	736	598	332	171	(10)
	CDD65	1553	0	1	3	11	65	220	379	424	286	145	18	1	(11)
	CDH74	13674	2	7	52	109	426	1719	3790	4382	2187	871	118	11	(12)
	CDH80	4667	0	0	13	31	117	502	1402	1719	622	239	21	1	(13)
(14) Wind	WSAvg	8.3	8.5	8.7	8.3	8.2	8.0	8.4	8.8	8.5	8.0	7.4	8.1	8.2	(14)
(15) Precipitation	PrecAvg	25.0	3.7	3.1	2.7	2.0	1.3	0.5	0.1	0.4	1.4	2.6	3.3	3.9	(15)
	PrecMax	44.4	8.9	7.4	8.8	6.9	3.5	2.0	0.7	2.6	3.5	10.7	9.3	9.1	(16)
	PrecMin	16.1	0.6	0.5	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2	0.3	(17)
	PrecStd	5.7	2.1	1.7	1.7	1.4	0.8	0.5	0.2	0.5	0.9	2.0	2.3	2.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.8	73.6	80.2	84.3	91.5	97.0	100.7	101.5	96.5	93.5	82.7	75.1	(19)
		MCWB	57.8	58.5	63.1	63.6	69.5	71.6	74.2	71.9	71.0	66.5	62.1	59.3	(20)
	2%	DB	66.7	68.2	73.3	76.5	82.0	89.4	93.3	94.7	89.6	85.9	77.1	69.5	(21)
		MCWB	56.4	57.1	60.6	63.6	67.3	72.1	73.3	74.4	72.3	68.2	61.7	57.9	(22)
	5%	DB	63.7	64.6	68.8	71.8	77.8	85.2	89.4	90.6	85.8	81.2	73.3	66.2	(23)
		MCWB	55.3	55.7	59.0	61.5	66.4	71.1	74.0	75.1	73.1	68.7	61.8	56.9	(24)
	10%	DB	61.0	61.2	65.7	69.4	75.2	82.2	86.3	87.6	82.8	78.6	70.0	63.6	(25)
		MCWB	54.2	54.4	57.5	60.8	65.4	70.5	74.3	75.8	72.5	68.7	61.3	56.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	61.1	64.8	68.1	72.3	77.5	80.4	81.8	79.2	75.7	69.1	61.8	(27)
		MCDB	66.8	68.6	75.1	79.1	83.9	85.9	86.3	87.0	84.5	80.3	73.7	69.4	(28)
	2%	WB	57.9	58.5	62.0	64.9	69.6	75.0	78.7	79.5	77.1	73.0	66.1	59.8	(29)
		MCDB	64.1	65.7	69.6	72.8	77.5	83.3	85.5	85.9	83.0	78.2	71.7	66.3	(30)
	5%	WB	56.4	56.6	60.0	63.3	67.9	73.3	77.2	78.1	75.7	71.2	63.9	58.3	(31)
		MCDB	62.0	62.9	66.8	70.0	75.2	81.1	84.4	85.0	81.8	77.2	70.0	64.3	(32)
	10%	WB	54.9	55.1	58.5	61.7	66.6	71.9	75.9	76.9	74.2	69.4	62.0	56.9	(33)
		MCDB	60.2	60.8	64.5	67.6	73.8	79.6	83.5	84.2	80.6	76.1	67.9	62.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.7	16.5	17.6	18.2	18.6	18.7	18.9	18.4	17.4	18.5	16.9	16.1	(35)
		MCDBR	20.5	22.1	24.3	24.0	24.1	24.4	24.6	23.6	21.3	24.2	22.7	21.1	(36)
	5% WB	MCWBR	12.5	13.1	14.0	13.7	12.1	11.1	10.9	11.2	10.2	11.8	12.3	12.4	(37)
		MCWBR	18.4	19.8	21.5	21.0	20.9	20.3	19.7	19.5	18.3	20.3	19.2	18.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.347	0.366	0.418	0.457	0.480	0.459	0.491	0.479	0.465	0.426	0.383	0.354	(40)
		taud	2.452	2.393	2.230	2.141	2.107	2.205	2.116	2.150	2.201	2.274	2.365	2.435	(41)
	Ebn at Noon		266	274	268	263	258	263	253	254	250	248	248	254	(42)
		Edh at Noon	27	32	42	48	50	46	49	47	43	36	30	26	(43)
(44) All-Sky Solar Radiation	RadAvg		723	978	1382	1730	2044	2370	2382	2091	1572	1189	810	655	(44)
	RadStd		78	90	115	131	129	72	68	90	75	76	56	60	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.35	N/A	+0.74	+0.85	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page